Lecture 17 // Oct 20 // Proof of the NST theorem

Last time:
We defined the r_s-invariant for $s \in [-\infty, 0]$

$$r_s(Y) = -\inf_{r<0} \{r \mid \exists\ \Theta\text{-supported } [r, -s] \text{ cycle}\}$$

which asks the question "in which filtered subquotients do we have a cycle of the form $\Theta + \dots$"

1. r_s-invariant is a piecewise constant, non-increasing function.
2. If there exists a negative-definite cobordism W (with $H_1(W) = 0$) from Y_1 to Y_2, then $r_s(Y_1) \leq r_s(Y_2)$. In particular, r_s is a homology cobordism invariant: if Y bounds a negative-definite 4-manifold W (with $H_1(W) = 0$), then $r_s(Y) = \infty$.
3. In addition, if $\pi_1(W) = 1$, then the inequality is strict (allowing $\infty \leq \infty$).

Today, we'll prove one of the main theorems of the NST paper.

Theorem 1 (NST)
Let $k \subseteq S^3$. Denote $Y_n = S^3_{1/n}(k)$, $n > 0$.

- If $[D_2^{Y_1}] \neq 0$, then none of the Y_n are homology cobordant to each other.
- If $r_s(Y_1) < \infty$ for any s, then $\{Y_n\}_{n=1}^{\infty}$ is linearly independent in $\Theta^3_{\mathbb{Z}}$.

Remark 2
A-priori, it's not obvious that $\Theta^3_{\mathbb{Z}}$ is at all interesting. For many years after its definition, it was not known whether $\Theta^3_{\mathbb{Z}}$ is 0, or ∞-generated. It is: in particular, it contains a $\mathbb{Z}^\infty$ subgroup ($\bigoplus_{i=1}^{\infty} \mathbb{Z}_i$). The original proof was due to Furuta and FS in the 80's. In some sense, this theorem is a generalization of theirs. Furuta and FS had the same family, and also used Yang-Mills theory. (Modern language (filtered) allows them to generalize it.)

Remark 3
It's actually very difficult to study $1/n$ surgeries using Heegaard Floer (HF). There's no known invariant coming from HF which distinguishes ∞-ly many $1/n$ surgeries on a fixed knot.

To study linear combinations of Y_n, we must first understand the behavior of r_s under $\#$.

Theorem 4
For any $s \in [-\infty, 0]$ and $s_1, s_2 \in [-\infty, 0]$ such that $s_1 + s_2 = s$,

$$r_s(Y_1 \# Y_2) \geq \min\{r_{s_1}(Y_1) + s_2,\ r_{s_2}(Y_2) + s_1\}.$$

Perhaps the form of this theorem – i.e., it is an inequality, and takes a minimum over a sum of r_s's and s's – is strange. We'll first give a pseudo-proof of the inequality, since the reason why it's of such a strange form will become apparent during the proof.

Proof (sketch). We'll pretend that $\underline{C}(Y_1\#Y_2) = \underline{C}(Y_1) \otimes \underline{C}(Y_2)$. This (# corresponds to $\otimes$) is the behavior one usually expects for complexes under connect sum.

The reducible in $\underline{C}(Y_1\#Y_2)$ is $\Theta_1 \otimes \Theta_2$, and $\widetilde{\mathrm{CS}}(\alpha \otimes \beta) = \widetilde{\mathrm{CS}}(\alpha) + \widetilde{\mathrm{CS}}(\beta)$.

Now, fix s_1 and s_2. Suppose that we have a Θ-supported cycle (a "OSC") α for $\underline{C}(Y_1)$ in $[r_1, -s_1]$, and a Θ-supported cycle β for $\underline{C}(Y_2)$ in $[r_2, -s_2]$.

What "Θ-supported" means is that in $\underline{C}(Y_1)$, $\alpha = \Theta + \dots$. To say that α is a cycle in this subquotient is to say that (a) α is in the subquotient, i.e., $\widetilde{\mathrm{CS}}(\alpha) \leq -s$, and (b) $\widetilde{\mathrm{CS}}(\partial\alpha) < r_1$. In particular, $\partial\alpha$ need not be 0, but if it isn't, it's outside the subquotient. The same is true for β: $\widetilde{\mathrm{CS}}(\beta) \leq -s_2$ and $\widetilde{\mathrm{CS}}(\partial\beta) < r_2$.

Claim: $\alpha \otimes \beta$ is a $\Theta_1 \otimes \Theta_2$-supported cycle (a.k.a., Θ-supported in the $\otimes$).

We know that $\widetilde{\mathrm{CS}}(\alpha \otimes \beta) \leq (-s_1) + (-s_2) = -s$. So $\alpha \otimes \beta$ is in the subquotient. By the Leibniz rule,

$$\partial(\alpha \otimes \beta) = (\partial\alpha) \otimes \beta + \alpha \otimes (\partial\beta).$$

Individually, $(\partial\alpha) \otimes \beta < r_1 + (-s_2)$ and $\alpha \otimes (\partial\beta) < (-s_1) + r_2$. Therefore $\alpha \otimes \beta$ is a cycle in the subquotient $[\max\{r_1 + (-s_2),\ (-s_1) + r_2\},\ -s]$. This is where the formula appears.

Now, note that this is true for all Θ-supported cycles in these subquotients. When does this occur? By definition, $-r_{s_1}(Y_1)$ and $-r_{s_2}(Y_2)$ are the borders of when there exists a Θ-supported cycle in each respective subquotient. So the claim is that you can take any r_1 up to $-r_{s_1}(Y_1)$ and any r_2 up to $-r_{s_2}(Y_2)$. Thus we get a Θ-supported cycle in the subquotient

$$[\max\{-r_{s_1}(Y_1) + (-s_2),\ -r_{s_2}(Y_2) + (-s_1)\},\ -s]$$

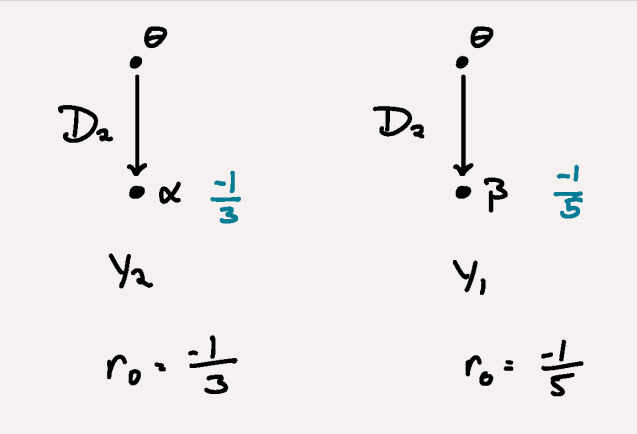
$$\Rightarrow\ r_s(Y_1\#Y_2) \geq \max\{r_{s_1}(Y_1) + s_2,\ r_{s_2}(Y_2) + s_1\}. \qquad \square$$

Corollary 5
$r_0(Y_1\#Y_2) \geq \min\{r_0(Y_1), r_0(Y_2)\}$.

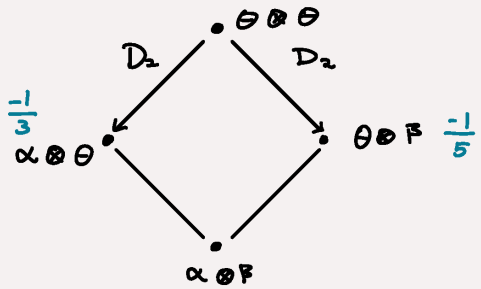
To see why this behavior – taking a minimum instead of adding – is more natural in this setting, let's do an example.

Example 6

Let the following be the $\underline{C}$ complexes for Y_1 and Y_2.



Here, r_0 is measuring the CS displacement of the D_2 arrow. The point is that if we pretend # is $\otimes$, the complex for $Y_1 \# Y_2$ looks like:



Notice that $D_2(\Theta)$ has 2 arrows, corresponding to the 2 summands in the Leibniz rule. Here, it's clear that to kill all the D_2's, you must take the minima of all of the summands.

Unfortunately, it's not actually true, in general, that $\underline{C}(Y_1 \# Y_2) = \underline{C}(Y_1) \otimes \underline{C}(Y_2)$. The rough reason is that it turns out that $\underline{C}(Y_1 \# Y_2)$ cannot be determined from $\underline{C}(Y_1)$ and $\underline{C}(Y_2)$ alone. To see why, let's look at the potential types of generators of $\underline{C}(Y_1 \# Y_2)$:

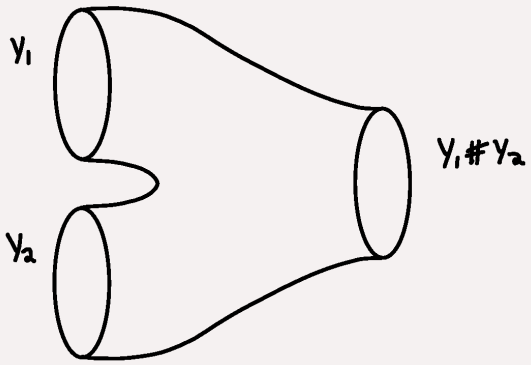
- $\Theta \otimes \Theta$ (red $\otimes$ red): this is red,
- $\alpha \otimes \beta$ (irred $\otimes$ irred),
- $\Theta \otimes \beta$ (red $\otimes$ irred),
- $\alpha \otimes \Theta$ (irred $\otimes$ red): these last three are irreducible in the tensor product.

The reason why the last two, which are combinations of reducible and irreducible generators, are irreducible in the product is as follows. Recall that generators correspond to SU(2) representations, and that $\pi_1(Y_1 \# Y_2) \cong \pi_1(Y_1) * \pi_1(Y_2)$. If we have SU(2) representations ρ_1 and ρ_2 of $\pi_1(Y_1)$ and $\pi_1(Y_2)$, and ρ_1 has trivial centralizer but ρ_2 doesn't, then $\rho_1 * \rho_2$ also has trivial centralizer. The intuition here is the same as for gauge transformations on the composite cylinder: even if some element commutes with ρ_2, it doesn't with ρ_1, so it doesn't commute with $\rho_1 * \rho_2$ either.

The issue arises because $\alpha \otimes \Theta$ and $\Theta \otimes \beta$ are irreducible: there is no reason to expect that there are no flows into/between them! In particular, we should be able to have maps from $\alpha \otimes \beta$ to $\alpha \otimes \Theta$. However, even if there is a D_1 map where $D_1(\beta) = \Theta$, in choosing the $\underline{C}$ complex, we threw these out! So you actually need both the D_1 and D_2 maps of each factor to calculate the $\underline{C}$ complex of the connect sum.

Remark 7
Being able to write down an explicit formula for $\underline{C}(Y_1 \# Y_2)$ as some tensor product is one motivation for why one may want to try harder to incorporate both D_1 and D_2 into the differential.

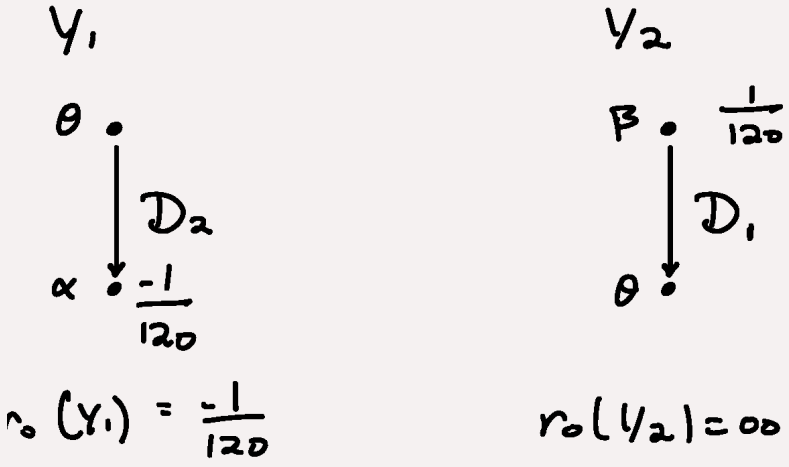
What salvages this is the fact that there is a cobordism map $\underline{C}(Y_1) \otimes \underline{C}(Y_2) \to \underline{C}(Y_1 \# Y_2)$. This cobordism is literally the pair of pants, from $Y_1 \sqcup Y_2$ to $Y_1 \# Y_2$.



Thus far, we've assumed that the Y's are connected, which seems silly to note, but the distinction between connected and disconnected boundary really matters to the gauge group: they're qualitatively different. It turns out that we get bordism maps from disconnected $\to$ connected, but not the other way. But this means that the r_s invariant of the algebraic complex $\underline{C}(Y_1) \otimes \underline{C}(Y_2)$ bounds the r_s invariant of the complex we actually care about, $\underline{C}(Y_1 \# Y_2)$.

Question 8. *Is the inequality always strict?*

Answer. No! For example, take $Y_1 = -\Sigma(2,3,5)$ and $Y_2 = \Sigma(2,3,5)$. Then $Y_1 \# Y_2$ is homology cobordant to S^3. The $\underline{C}$ complex for each summand looks like



Since r_s is a homology cobordism invariant,

$$\infty = r_0(S^3) = r_0(Y\# - Y) \geq \min\{r_0(Y), r_0(-Y)\} \geq \min\left\{\frac{1}{120}, \infty\right\} \geq \frac{1}{120}.$$

We are now prepared to prove the theorem. We'll prove a more general algebraic statement.

Theorem 9
Let $\{Y_n\}_{n=1}^{\infty}$ be a sequence of $\mathbb{Z}\text{HS}^3$'s such that

1. $\infty > r_0(Y_1) > r_0(Y_2) > \dots$
2. $\infty = r_0(-Y_1) = r_0(-Y_2) = \dots$

Then the $\{Y_n\}_{n=1}^{\infty}$ are linearly independent in $\Theta^3_{\mathbb{Z}}$.

Proof. Assume not: suppose $\sum_{i\leq m} n_i Y_i = 0$ in $\Theta^3_{\mathbb{Z}}$. Then

$$n_m Y_m = \sum_{i<m} n_i Y_i \qquad \text{(assume } n_m > 0\text{)},$$

so $r_0(n_m Y_m) = r_0\left(\sum_{i<m} n_i Y_i\right)$. Note that these n_i can be positive or negative. Iterating this inequality, we get that

$$r_0(n_m Y_m) = \dots \geq \min\{r_0(\pm Y_1), r_0(\pm Y_2), \dots\},$$

where the $+$ or $-$ in each $\pm$ is chosen based on $\text{sign}(n_i)$.

On the other hand, since $Y_m = n_m Y_m \#(-n_{m-1})Y_m$,

$$r_0(Y_m) = r_0\left(n_m Y_m \#(-n_{m-1})Y_m\right) \geq \min\{r_0(n_m Y_m), r_0(-Y_m)\} \geq \min\{r_0(n_m Y_m), \infty\} = r_0(n_m Y_m),$$

since $r_0(-Y_m) = \infty$ by hypothesis. Thus $r_0(Y_m) \geq r_0(n_m Y_m)$, so

$$r_0(Y_m) \geq \min\{r_0(\pm Y_1), r_0(\pm Y_2), \dots\},$$

which contradicts the initial hypotheses. □

The approximate intuition for this result is that small and nonzero CS displacement of D_2 cannot lie in the span of large CS displacements.

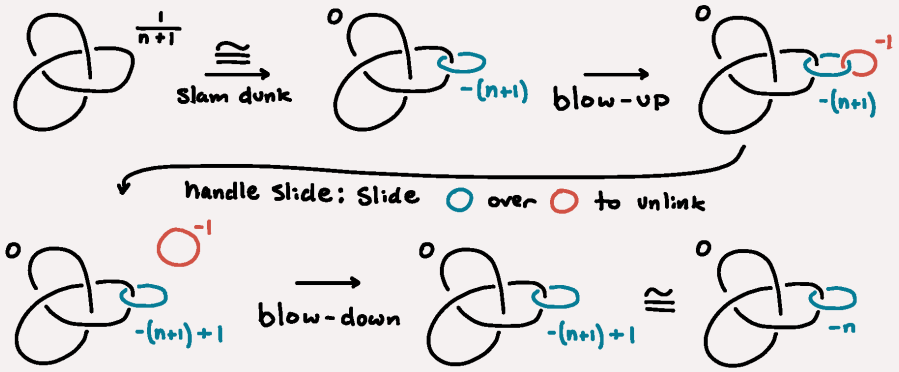
Does this apply to our situation? Yes! Here's a slick topological argument, which concludes the proof of NST.

Theorem 10 (Nozaki-Sato-Taniguchi)
Let $k \subseteq S^3$. Denote $Y_n = S^3_{1/n}(k)$, $n > 0$. If $[D_2^{Y_1}] \neq 0$, then none of the Y_n are homology cobordant to each other.

Proof. We only need to show that the conditions of the previous theorem are satisfied. E.g., we need to verify that

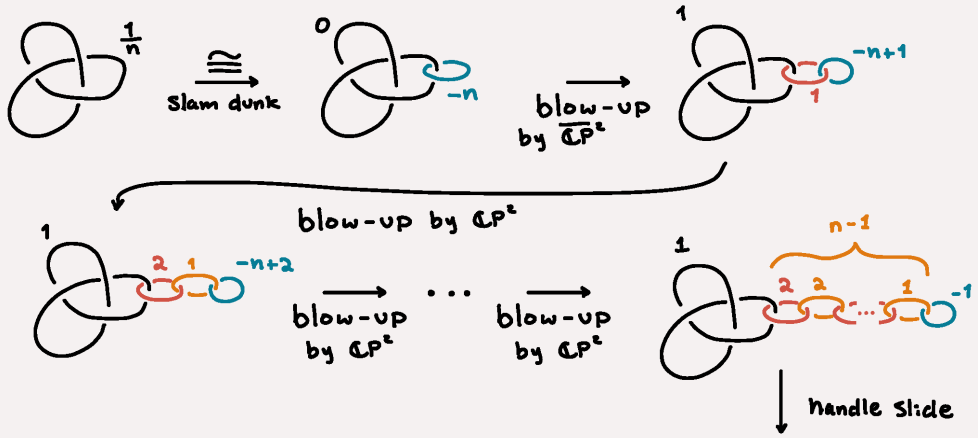
1. $\infty > r_0(Y_1) > r_0(Y_2) > \dots$
2. $\infty = r_0(-Y_1) = r_0(-Y_2) = \dots$

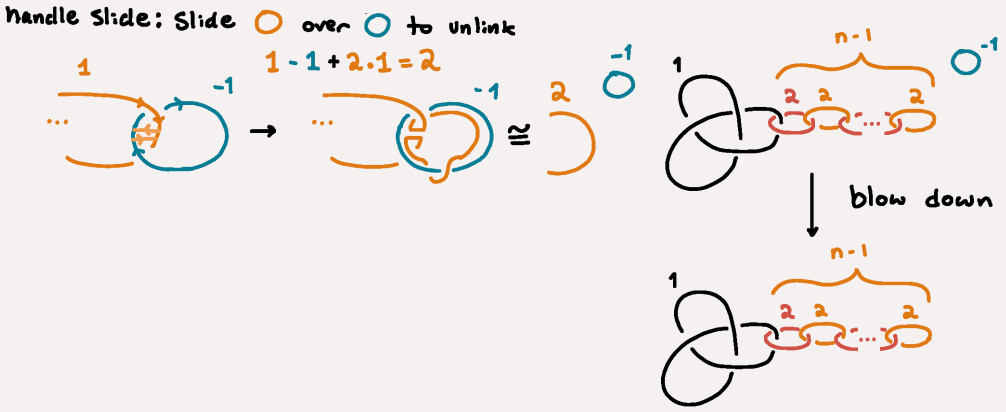
(1) holds from last time: we constructed a $\pi_1 = 1$ negative-definite cobordism from Y_{n+1} to Y_n:



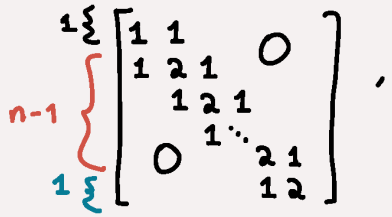
(2) follows from the fact that $S^3_{1/n}(k)$ bounds a positive-definite manifold if $n > 0$ and a negative-definite manifold if $n < 0$.

The positive-definite manifold can be constructed as follows ($n > 0$):





The intersection form is



which we leave to the reader to verify has Smith normal form Id.

The same process can be applied when $n < 0$ by blowing up with $\overline{\mathbb{CP}}^2$'s. Since this process gives a negative-definite cobordism from $-Y_n$ to S^3, we can conclude that $r_0(-Y_n) = \infty$. □